[인공지능]

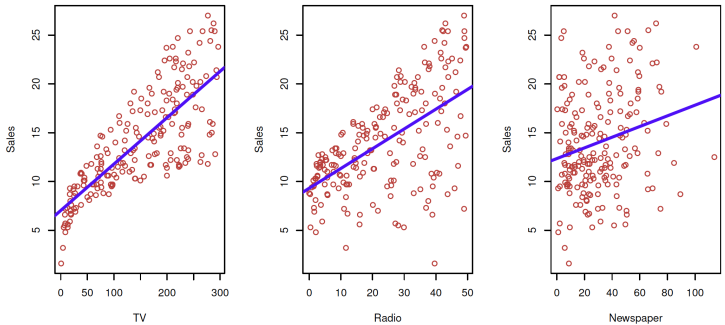
Regression

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Advertising Data

- Contains data on the budget allocated for TV, radio and newspaper advertisements with the resulting sales.



Questions from Data

- Questions we might ask:
  - Is there a relationship between advertising budget and sales?
  - How strong is the relationship between advertising budget and sales?
  - Which media contribute to sales?
  - How accurately can we predict future sales?
  - Is the relationship linear?
  - Is there synergy among the advertising media?

→ **Regression analysis** can be used to estimate it.

Terminology

- Independent variable $x, \mathbf{x}_i, X, \mathbf{X}$
  - Input
  - Feature
  - Attribute
  - Predictor
  - Risk factor
- Dependent variable $y, \mathbf{y}_i, Y, \mathbf{Y}$
  - Output
  - Target
  - Label
  - Response
  - Class
  - Outcome

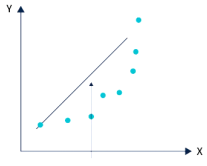
What is Regression?

- **Linear Regression Model** represents the output variable Y as a linear combination of the input variables X.
  - Linear combination is a form where several variables are multiplied by coefficients and then added together.
- Objective for linear regression
  - Explain the relationship between variable X and variable Y in mathematical form.
  - Predict the future value of the response variable Y.

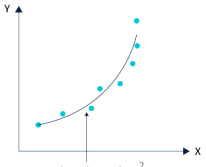
Category of Regression

- **Simple regression**: regression with a single independent variable
  - Linear
  - Nonlinear

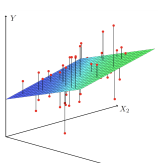
Simple Linear Model



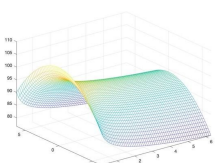
Polynomial Model



- **Multiple regression**: regression with two or more independent variables
  - Linear
  - Nonlinear

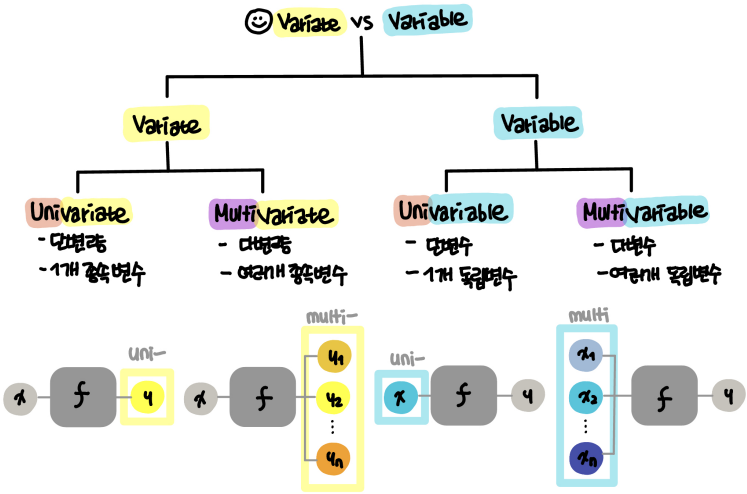


Multiple linear regression



Multiple nonlinear regression

Multiple (Multivariable) vs. Multivariate



Simple Linear Regression

- We assume a model

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where β_0 and β_1 are two unknown constants that represent the **intercept** and **slope**, also known as **coefficients** or **parameters**, and ϵ is the **error** term.

- Given some estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for the model coefficients, we predict future sales using

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x,$$

where $\hat{y}$ indicates a prediction of Y on the basis of $X = x$.
The *hat* symbol denotes an estimated value.

Estimation of the Parameters by Least Squares

- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be the prediction for Y based on the i-th value of X.
- **Residual**: $e_i = y_i - \hat{y}_i$
- **Residual Sum of Squares (RSS)** as

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2,$$

or equivalently as

$$\text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \cdots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2.$$

$\rightarrow$ Find estimator of β_0 and β_1 (*i.e.*, $\hat{\beta}_0$ and $\hat{\beta}_1$).

Least Squares Estimation Algorithm

Step 1.

$$\sum_{i=1}^{n} \{y_i - (\beta_0 + \beta_1 x_i)\}^2$$

Step 2.

$$C(\beta_0, \beta_1) = \min_{\beta_0, \beta_1} \sum_{i=1}^{n} \{y_i - (\beta_0 + \beta_1 x_i)\}^2$$

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{\beta_0, \beta_1} C(\beta_0, \beta_1)$$

Step 3.

$$\frac{\partial C(\beta_0, \beta_1)}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i)) = 0$$

$$\frac{\partial C(\beta_0, \beta_1)}{\partial \beta_1} = -2 \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i)) x_i = 0$$

Least Squares Estimation Algorithm Solution

Solution

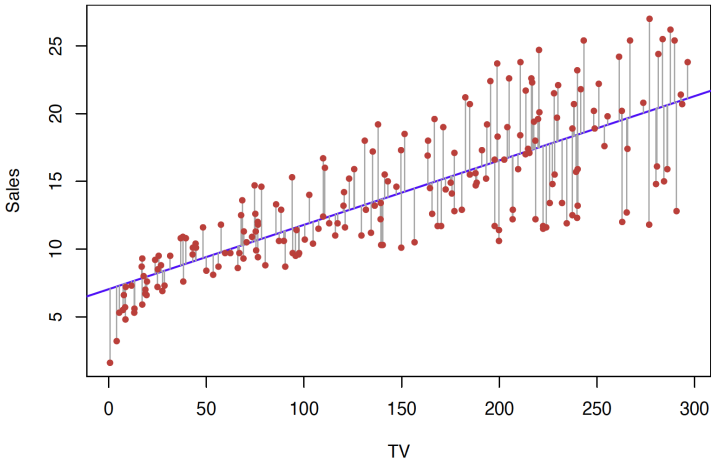
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

Example: Advertising Data

- The least squares fit for the regression of sales onto TV.

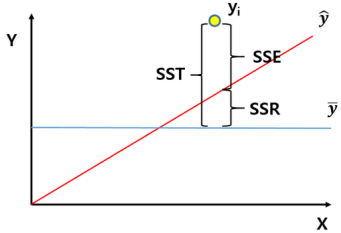


What is Coefficient of Determination, R^2?

- A measure of how well the regression model explains the variation in the data.
- Indicates the proportion of the total variance in y explained by the model.
- Range: $0 \leq R^2 \leq 1$ (sometimes negative if the model is very poor).
  - $R^2 = 1$: Good fit
  - $R^2 = 0$: Poor fit
- **Definition**

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Coefficient of Determination, R^2



SST(Y의 전체 변동) : $\sum(y_i - \bar{y})^2$
SSR(모형에 의해 설명되는 변동) : $\sum(\hat{y}_i - \bar{y})^2$
SSE(모형에 의해 설명이 되지 않는 변동) : $\sum(y_i - \hat{y}_i)^2$

Source

- The total variation of y can be decomposed into:

$$SST = SSR + SSE$$

SST (Total Sum of Squares)

$$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

SSR (Regression Sum of Squares)

$$SSR = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$$

SSE (Error Sum of Squares)

$$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

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Multiple Linear Regression

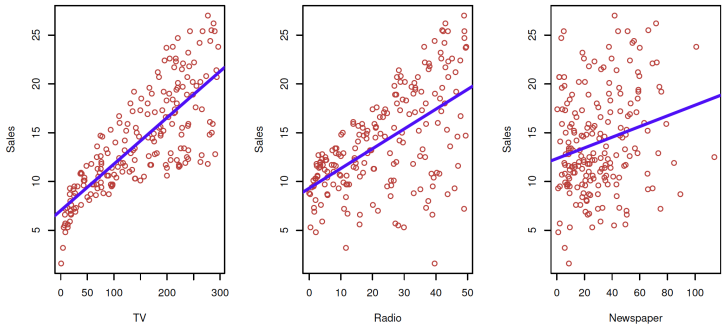
- Our model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

- We interpret β_j as the *average* effect on Y of a one unit increase in X_j, holding all other predictors fixed.

- In the advertising example, the model becomes

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon.$$



Estimation and Prediction for Multiple Regression

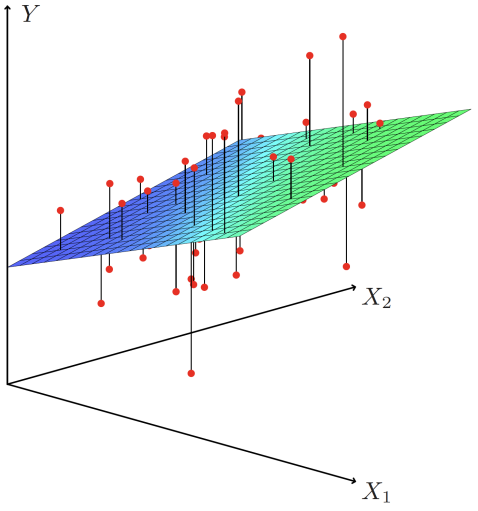
- Given estimates $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$, we can make predictions using the formula

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p.$$

- We estimate $\beta_0, \beta_1, \ldots, \beta_p$ as the values that minimize the sum of squared residuals

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \cdots - \hat{\beta}_p x_{ip})^2. \end{aligned}$$

The values $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ that minimize RSS are the multiple least squares regression coefficient estimates.



Multiple linear regression

Adjusted R^2

- **Definition**

$$\text{Adjusted } R^2 = 1 - \frac{SSE/(n-p-1)}{SST/(n-1)}$$

where

  - R^2: Coefficient of determination
  - n: Sample size
  - p: Number of predictors

- **Interpretation:**
  - The adjustment term $(n-1)/(n-p-1)$ penalizes the addition of predictors
  - Prevents overestimation of model fit when more predictors are added

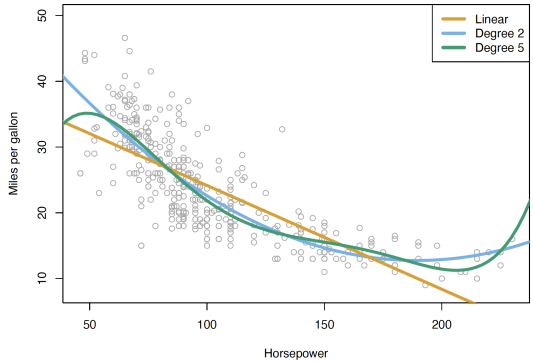
Interpreting Regression Coefficients

- The ideal scenario is when the predictors are uncorrelated — a *balanced design*:
  - Each coefficient can be estimated and tested separately.
  - Interpretations such as *"a unit change in X_j is associated with a β_j change in Y, while all the other variables stay fixed"*, are possible.
- Correlations among predictors cause problems:
  - The variance of all coefficients tends to increase, sometimes dramatically
  - Interpretations become hazardous — when X_j changes, everything else changes.
- *Claims of causality* should be avoided for observational data.

Polynomial Regression

- Non-linear effects of predictors

$$\texttt{mpg} = \beta_0 + \beta_1 \times \texttt{horsepower} + \beta_2 \times \texttt{horsepower}^2 + \epsilon$$



Example of polynomial regression

Important Questions in Regression

1. Is at least one of the predictors $X_1, X_2, \ldots, X_p$ useful in predicting the response?

$\rightarrow$ t-test for individual regression coefficients, F-test for overall regression model

2. Do all the predictors help to explain Y, or is only a subset of the predictors useful?

$\rightarrow$ Forward/Backward selection

3. How well does the model fit the data?

$\rightarrow$ R^2, Adjusted R^2

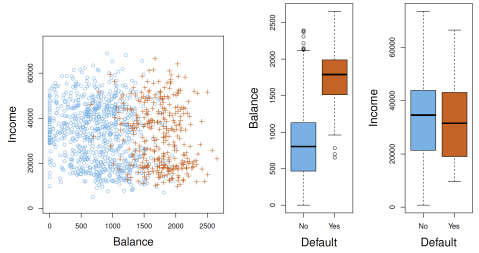
4. Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

$\rightarrow$ Confidence interval

Can We Use Linear Regression for Classification?

- Suppose for a classification task,

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes} \end{cases}$$



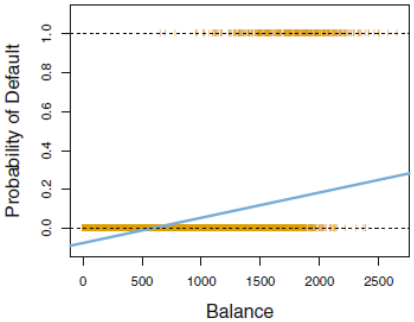
- Can we simply perform a linear regression of Y on X and classify as Yes if $\hat{Y} > 0.5$?

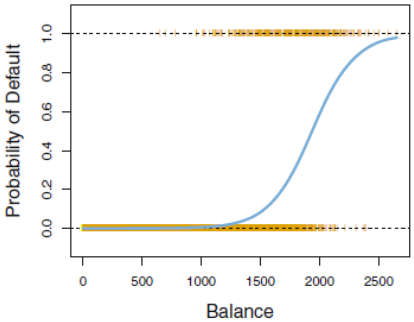
→ Since in the population $E(Y|X = x) = Pr(Y = 1|X = x)$, we might think that regression is perfect for this task.

- However, linear regression might produce probabilities less than zero or bigger than one.

→ **Logistic Regression** is more appropriate.

Linear vs. Logistic Regression





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Logistic Regression

- Let's write $p(X) = \Pr(Y = 1 \mid X)$ for short and consider using balance to predict default.
- Logistic regression uses the form

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}} = \frac{1}{1 + e^{-(\beta_0+\beta_1 X)}}.$$

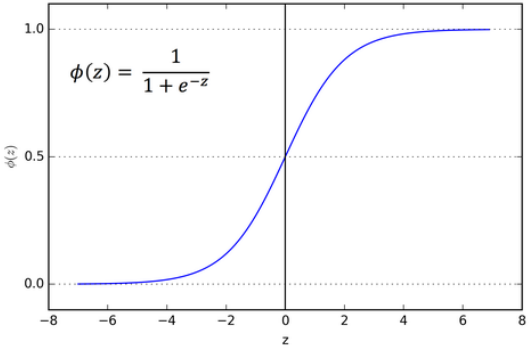
  - $p(X)$ will have values between 0 and 1.

- A bit of rearrangement gives:

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X.$$

This transformation is called the **log odds** or **logit** transformation of $p(X)$.

Sigmoid Function



Maximum Likelihood

- Use maximum likelihood to estimate the parameters.

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

- Example What is our estimated probability of default for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1000}}{1 + e^{-10.6513+0.0055\times 1000}} = 0.006$$

With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 2000}}{1 + e^{-10.6513+0.0055\times 2000}} = 0.586$$

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Logistic Regression for Classification

- **Key idea:** Logistic regression outputs a probability

$$\hat{p}(X) = P(Y = 1 \mid X).$$

- **Classification rule:**

$$\text{Predict } Y = \begin{cases} 1 \text{ (Yes, Default)} & \text{if } \hat{p}(X) > 0.5, \\ 0 \text{ (No, Non-default)} & \text{if } \hat{p}(X) \leq 0.5. \end{cases}$$

- Example
  - Balance = 1000: $\hat{p}(X) = 0.006$ $\Rightarrow$ Predict No Default
  - Balance = 2000: $\hat{p}(X) = 0.586$ $\Rightarrow$ Predict Default

$\rightarrow$ Logistic regression is not just about estimating probabilities, but also about making classification decisions.

Thank You!

(Q & A)

References

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